

Guowei Huai

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Research Interests

Dexterous manipulation and real-world robot learning, with a focus on using human data for robot policy pre-training and post-training, including fine-tuning for real-world tasks. Related interests include contact-rich tool use, in-hand manipulation, teleoperation, and tactile sensing.

Education

HKUST (Guangzhou)

PhD student, Robotics and Autonomous Systems

Academic Advisor: Jie Song

Sept. 2026 – Present

Guangzhou, China

HKUST (Guangzhou)

MPhil, Robotics and Autonomous Systems

Red Bird MPhil (RBM) Program

Academic Advisor: Jie Song | Project Advisor: Arthur Kar Leung Lin

GPA: 4.04/4.30

Sept. 2024 – Sept. 2026

Guangzhou, China

Beijing Institute of Technology, Zhuhai

B.Eng., Computer Science

GPA: 89.8/100

Sept. 2019 – June 2023

Zhuhai, China

Publications

HiFun: A Hierarchical Framework for Efficient Functional Dexterous Manipulation Learning

Linyi Huang, **Guowei Huai**, Weibin Liu, Shulong Jiang, Ping Tan, Weixuan Zhang, Hui Zhang, Jie Song.

Conference on Robot Learning (CoRL), 2026. Accepted. Project Paper

Research Experience

HiFun: Functional Dexterous Manipulation

HKUST (Guangzhou), HKUST & ETH Zurich | Core contributor; second author

May 2025 – May 2026

- Contributed to a hierarchical real-world RL system that separates contact-aware hand skills from arm control and skill activation for functional tool use.
- Evaluated BC, diffusion policy, human-in-the-loop RL, retargeting, and replay baselines on Franka Research 3 with XHand and Sharpa; analyzed contact failures, timing errors, and robustness under perturbations.
- The complete framework achieved 98.3% mean success across six tasks and two hands, with at most 60 minutes of online HIL training per task; the full pipeline averaged approximately 109 minutes per task.

Research Experience

Goal-Conditioned In-Hand Object Reorientation

Dec. 2025 – Present

HKUST (Guangzhou) | MPhil thesis project

- Developed a goal-conditioned in-hand rotation framework for object reorientation, including elongated objects, with a desired object pose as the control goal.
- Formulated any-grasp-to-any-grasp transitions: reorient an object from an initial grasp toward a target grasp by conditioning the policy on the target object pose.
- Built a two-stage reinforcement-learning pipeline in IsaacLab, learning force-closure grasps before multi-axis object rotation with tactile feedback on Sharpa Hand.

Dexterous Teleoperation and Multimodal Data Platform

May 2025 – Mar. 2026

HKUST (Guangzhou) | Project leader

- Developed optimization-based retargeting for Manus gloves and Vision Pro using vector and fingertip objectives; unified both inputs in one repository with a shared interface for Inspire Hand, Leap Hand, XHand, and Sharpa.
- Integrated GELLO arm control with Manus hand retargeting for coordinated Franka arm–hand demonstrations; also supported a 3D mouse and Vision Pro end-effector control.
- Integrated piezoresistive, electromagnetic, and vision-based tactile sensors; synchronized tactile streams, robot states, and demonstrations for real-world RL and imitation learning.

Diffusion-Based Visual Imitation Learning

Mar. 2025 – May 2025

HKUST (Guangzhou) course project | Contributor Code Report

- Reproduced DP3 and RDT-1B on AIRBOT Play; processed 490 Airbot500 teleoperation episodes through workspace cropping, farthest-point sampling, and Zarr conversion.
- Analyzed real-robot evaluations reporting 72% seen-task success and 30% unseen-container generalization.

Teaching & Service

Reviewer, ICRA Workshop on Manipulation Robustness

May 2026

Course instructor & designer, Datawhale

May 2023 – Sept. 2024

Designed AI4S competition courses, built baselines for beginners, and led online teaching sessions.

Selected Honors

First Prize, Bionic Robot Innovation Competition

Oct. 2025

HKUST (Guangzhou); MoSense team

Quarterfinalist (Top 8), 1st WBCD Competition at ICRA

May 2025

Full Scholarship, HKUST (Guangzhou) MPhil RBM Program

Sept. 2024

Principal's First-Class Scholarship, BITZH; 1/2000

Apr. 2023

Technical Skills

Robot learning: reinforcement and imitation learning, diffusion policy, PyTorch, JAX.

Robotics systems: ROS, Isaac Gym/Lab, dexterous retargeting, tactile sensing, multimodal data collection.

Development: Git workflows, Claude Code, Codex.